**21** Sketch the graph of

$$y = 2x^2 - 8x - 5$$

showing the coordinates of the turning point and the exact coordinates of any intercepts with the coordinate axes.

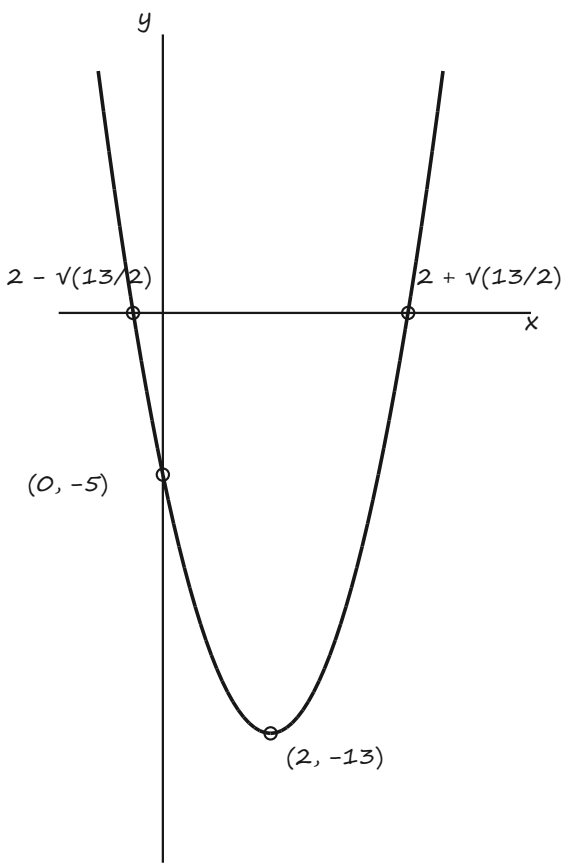
$y = 2x^2 - 8x - 5 = 2(x - 2)^2 - 13.$

Turning point: $(2, -13)$; $y$-intercept: $(0, -5)$.

For $y = 0$, $(x - 2)^2 = 13/2$,

so $x$-intercepts: $(2 \pm \sqrt{(13/2)}, 0)$.

**5 marks total**
Complete square / start roots
Exact roots
Turning point
y-intercept
Correct labelled sketch




**(Total for Question 21 is 5 marks)**